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Project #034: Collaborative Approach to Length of Stay Reduction in Complex Patients

Katherine R. Flannery
Henry Ford Health, KFLANNE1@hfhs.org

Jim McAlpine
Henry Ford Health, jmcalpi1@hfhs.org

Hannah Musgrove
Henry Ford Health, hmusgro1@hfhs.org

Phina Ross
Henry Ford Health, pross5@hfhs.org

Margot Tilley
Henry Ford Health, mtilley5@hfhs.org

See next page for additional authors

Authors
Katherine R. Flannery, Jim McAlpine, Hannah Musgrove, Phina Ross, Margot Tilley, Romina Bello, Jonathan Vono, Bruce Leaman, Nancy Price, and Neil Khanna

Collaborative Approach to Length of Stay Reduction in Complex Patients

HENRY FORD HEALTH

WINNER #4

Katherine Rose Flannery, MD, Associate Chief Medical Officer, Hospital Administration; Jim McAlpine, Principal Management Engineer, Quality & Safety; Hannah Musgrove, MSN, APRN, AGCNS-BC, Principal Management Engineer, Quality & Safety; Phina Ross, MHA, MSW, LMSW, Director, System Inpatient Care Coordination & Patient Safety; Margot Tilley, MHA, RN, ACMA-RN Manager, Case Management; Romina Bello, Supervisor, Rehabilitation Services; Jonathon Vono, MBA, RRT, Clinical Manager, Respiratory Therapy Department; Bruce Leaman, System Vice President, Risk Finance & Insurance Services; Nancy Price, MSN, RN, NE-BC, Nursing Administrator, Patient Care Services; & Neil Khanna, MD, Medical Director of Utilization Management, Hospital Administration

Henry Ford Health, Detroit, Michigan

Abstract

Plan/Introduction: Patients with a length of stay (LOS) over 29 days are considered long-stay and often require complex discharge planning due to higher risks of poor outcomes.[1-2] To address this, Complex Care Rounds was previously launched to support care coordination and expedite discharge through weekly team meetings, which shifted to a virtual format during the pandemic.[3] Despite these efforts, the number of long-stay patients continued to rise in 2022. The goal of this quality improvement project was to enhance Complex Care Rounds to reduce the average LOS and number of discharges for long-stay patients.

Do/Methods: Guided by the Plan-Do-Study-Act cycle and the SQUIRE 2.0 framework, a multidisciplinary team evaluated Complex Care Rounds to identify opportunities for improvement.[4] The planning team renewed engagement by expanding representation from additional disciplines. Case management leadership was required to participate in Complex Care Rounds to provide real-time updates and facilitate open discussion of discharge barriers and solutions. A hospital policy was developed to provide guidance to care teams when patients or their surrogate decision-makers decline a reasonable discharge option.

Check/Results: The average LOS for long-stay patients has decreased by 7%, resulting in a reduction of 8,281 days. Additionally, the total number of long-stay patients has decreased by 18%, resulting in increased capacity for admitting additional patients.

Act/Discussion: Using a team-based approach to address the discharge barriers of long-stay patients proved effective in reducing LOS by fostering better communication, enabling timely problem-solving, and coordinating care across disciplines.

Plan / Introduction

Problem Description: Patients with a length of stay (LOS) greater than 29 days are categorized as long-stay patients at Henry Ford Hospital. These patients present complex discharge planning. For patients, prolonged hospitalization can lead to poorer outcomes and a reduced quality of life.[1-2]

Available Knowledge: Complex Care Rounds was established in 2016 to support the care of long-stay, complex patients, and facilitate earlier hospital discharge. Initially formed by a small core team, the group met weekly in person to review reasons for ongoing hospitalization and address discharge barriers through collaborative problem-solving. However, in 2022, the number of patients with a length of stay (LOS) greater than 29 days continued to increase, despite the ongoing implementation of Complex Care Rounds.

Rationale: Discharge delays occur when patients remain hospitalized, often leading to increased risk of adverse events such as infections, falls, and functional decline. Prolonged stays also heighten the chance of hospital-related complications. Preventing discharge delays helps avoid the negative outcomes associated with extended hospitalization.[3]

Goal: The goal of this quality improvement project was to enhance Complex Care Rounds to reduce the average LOS and number discharges for long-stay patients by 10% by the end of 2024.

Do / Methods

Guided by the Plan-Do-Study-Act cycle and the SQUIRE 2.0 framework, a multidisciplinary team evaluated Complex Care Rounds to identify opportunities for improvement.[4]

Interventions:

- Renew engagement with physicians and respiratory therapy
- Required participation by Case Management leadership
- Expanded representation to include physical therapy, care experience, and nursing administration
- Development of Declining Hospital Discharge Policy

Study of the Interventions: To assess the impact of Complex Care Rounds, the pre-intervention period (January – December 2022) was compared to the intervention period (January – December 2024).

Measures:

- Long-Stay Patients, Total Count
- Long-Stay Patients, Rate per 1,000 Patient Days
- Average LOS for Long-Stay Patients

Analysis: Data was tracked on the system dashboards. An observed decrease during the intervention period was considered a key indicator of success.

Check / Results

Project Modifications: Feedback from case management revealed that the original meeting day presented challenges for the team, promoting a shift to a new day that better aligned with their workflow. This change allowed Case Management to be more prepared to provide an informed update during the meeting. During development of the declining hospital discharge policy, the proposed process was presented to the Patient and Family Advisory Committee (PFAC) for feedback. PFAC expressed support for the policy and acknowledged its necessity. The planning team also consulted the legal department to ensure compliance and clarity prior to final policy approval (Figure 1). In addition to these changes, proactive reviews were conducted for patients with a length of stay exceeding 10 days beyond their geometric mean LOS. This allowed the teams to identify complex needs earlier.

Results: Since 2022, the average LOS for long-stay patients has decreased by 7%, resulting in a reduction of 8,281 days. Additionally, the total number of long-stay patients has decreased by 18%, resulting in increased capacity for admitting additional patients (Table 1).

Table 1: Complex Care Rounds Outcomes

Metric	2022	2024	Positive Change
Long-Stay Patients, Total Count	738	606	18% Reduction
Long-Stay Patients, Rate per 1000 Patient Days	3.52	2.72	23% Reduction
Average LOS for Long-Stay Patients	47.58	44.28	7% Reduction

Act / Discussion

Key Findings & Interpretation: Refining Complex Care Rounds has led to a reduction in LOS for long-stay patients by improving coordination and addressing discharge barriers more effectively. This reduction in hospital days helps minimize the risk of further complications, such as infections, functional decline, and other hospital-associated conditions that can arise from prolonged stays. The development of the declining hospital discharge policy provided staff with a clear, structured framework to manage and navigate the complex and sensitive issue of discharge refusal.

Spread: Since implementing the declining hospital discharge policy, the planning team has engaged with representatives from other hospitals within the system to share insights on the policy's development and implementation. The policy is planned to be a system policy in the future.

Conclusion: Using a team-based approach to address the discharge barriers of long-stay patients proved effective in reducing LOS by fostering better communication, enabling timely problem-solving, and coordinating care across disciplines. This collaborative strategy helped identify and overcome obstacles more efficiently, ultimately improving patient flow and outcomes while optimizing hospital resources.

Figure 1: QR Code to Declining Hospital Discharge Policy

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